

Madhulika Musku

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Highly skilled and motivated **Software Engineer** around **4 years** of experience in designing, developing, and maintaining robust web applications and data-driven solutions. Seeking to leverage expertise in front-end and back-end technologies, cloud services, and data analysis to contribute effectively to innovative projects in a dynamic environment.

EDUCATION:

University of Missouri- Kansas City, MO

Aug-2021 To Dec-2022

Master of Science in Computer Science

Jawaharlal Nehru Technological University, Hyderabad, India

Aug-2016 To Sep-2020

Bachelor of Technology in Computer Science and Engineering

TECHNICAL SKILLS:

Programming Languages: Java, JavaScript, Python, C#, C, C++.

Web Technologies: HTML5, CSS3, JavaScript, jQuery, TypeScript, Bootstrap, XHTML, XML, SOAP, REST, JSON.

Frameworks: ReactJS, Angular, NodeJS, ExpressJS, Django, Spring Boot, .NET.

Cloud Technologies: AWS, Azure, GCP.

Data Visualization: PowerBI.

IDE/Tools: IntelliJ, Visual Studio code, Eclipse, WebStorm.

Databases: MySQL, PostgreSQL, MongoDB, KQL.

Version Control Tools: GitHub, GitLab, Azure DevOps.

Operating Systems: Windows OS, Linux, Mac OS.

Containerization: Docker, Kubernetes.

EXPERIENCE:

The Mount Sinai Hospital, New York, NY

Jun-2024 To Present

Software Engineer

- Developed and implemented scalable Full-Stack applications using **JavaScript, React, Node.js** and **AWS**, improving patient portal accessibility and enhancing data processing efficiency for healthcare workflows.
- Integrated **RESTful APIs** and microservices to ensure seamless interaction between healthcare systems, achieving a **96.9%** uptime and optimizing response times for critical patient-facing features.
- Spearheaded service containerization with **Docker** and managed deployments using **AWS Elastic Kubernetes Service (EKS)**, reducing deployment times by **25%** and improving **CI/CD** pipeline efficiency.
- Enhanced data retrieval and storage by optimizing **MongoDB** and **PostgreSQL** queries, to support the scalability and high availability of patient and provider data during peak usage periods.
- Utilized **AWS** services such as **Lambda** and **EC2** to ensure efficient and reliable backend processing for dynamic healthcare workloads, including handling sensitive patient data securely.
- Streamlined version control and collaboration processes using **Git**, enabling the development team to deliver high-quality healthcare solutions efficiently and consistently.

Microsoft, Bellevue, WA

Nov-2023 To May-2024

Software Engineer

- Partnered with **Microsoft** to develop and manage robotic **KPIs**, ensuring accurate and insightful data analysis and visualization.
- Utilized **Kusto Query Language (KQL)** to efficiently query, analyze, and manipulate large datasets derived from robotic operations on **Azure Data Explorer**, optimizing data retrieval and processing.
- Designed and improved interactive **Power BI** dashboards for visualizing key performance indicators (KPIs), enabling data-driven decision-making. And identifying trends and generating actionable reports.
- Implemented sophisticated **KPI** UI dashboards using **C++** and **.Net** enhancing user experience and integrating seamlessly with other enterprise systems.

- Actively participated in **ROS (Robot Operating System)** planning and **KPI** implementation meetings, contributing to strategic discussions and technical planning.
- Developed machine learning models using TensorFlow to predict and optimize robotic operations, improving the accuracy and efficiency of automation processes.
- Managed and optimized continuous integration and continuous deployment (**CI/CD**) pipelines using **Azure DevOps**, ensuring reliable and efficient software delivery.
- Ensured data security and compliance with industry standards and best practices, implementing role-based access control and data encryption where necessary.

TypeEx Solutions, Irving, TX

Jul-2023 To Nov-2023

Software Engineer

- Participated in the development and integration of a high-performance web application leveraging **Java 8, Spring Boot, Hibernate, Angular 8, and MongoDB**. Ensured the delivery of a seamless and user-friendly interface.
- Utilized **AWS** for cloud deployment, skillfully utilizing services like **EC2, S3, and RDS**, while effectively integrating **MongoDB** for efficient data storage and retrieval.
- Implemented **Kubernetes** for container orchestration and **Jenkins** for continuous integration, optimizing deployment processes.
- Actively used **Git** for version control and collaborative development, ensuring efficient code management and seamless team collaboration.
- Administered **Linux** servers, performing tasks such as **Bash** scripting and system maintenance to ensure optimal performance and security.

Xenon Infotek, Edison, NJ

Jan-2023 To Jun-2023

Software Engineer

- Used **Java 8, Spring Boot, React, Node and MongoDB** for development of the application. Utilized **Azure** cloud services like **Azure App Service** and **Azure SQL Database** for integration and deployment for seamless, scalable and dependable application.
- Implemented **Kafka** for event-driven architectures and leveraged **Kubernetes** for containerizing resulting in real-time data processing and enhanced system resilience.
- Demonstrated expertise in unit testing with **JUnit**, ensuring the reliability and stability of software applications through comprehensive test coverage.
- Used **JSF (Java Server Faces)** for server-side UI component development, enhancing code reusability and maintainability.
- Applied **Terraform, Ansible, Chef, and Puppet** for infrastructure automation, ensuring consistent and repeatable deployments.

Bizongo, Hyderabad, India

Jun-2019 To Jul-2021

Software Engineer

- Expanded robust and scalable **Java/J2EE** applications using **Spring Boot, REST API, and PostgreSQL**, resulting in improved system performance and enhanced user experience.
- Used **Node.js** and **Express JS** to build efficient and lightweight server-side applications, enabling seamless handling of concurrent requests and high-performance data processing.
- Collaborated with cross-functional teams using **JIRA** for agile project management, facilitating effective communication, task tracking, and issue resolution throughout the development lifecycle.

ACADEMIC PROJECTS:

Grading System in Java: Developed the Grading System project using Java, HTML, CSS, and JavaScript. The application offers an intuitive user interface developed with Java Swing and HTML/CSS for the graphical elements. This interface allows teachers and administrators to input students' scores for assignments, exams, and quizzes. Applied the grading logic to calculate final grades based on predefined grade boundaries and assessment weights. This project includes data validation to ensure accurate inputs, achieved through JavaScript validation on the front-end and Java validation on the back-end. This ensures that users enter valid data within the specified range and generates grade reports for all students, displaying their final grades. I have used Java Swing for generating text-based reports on the console and HTML/CSS for visual reports in a separate web page.

Predictive Analysis for Determining Safety of a City (python, seaborn, Pandas, geoplot): A python-based application which is used to determine the safety of the city analyzing the available datasets considering various parameters like Crime rate, Pollution, Number of accidents in the city and display the results.

CERTIFICATIONS:

- **Full Stack Java Developer** certification from **Udemy**.
- **Master Microservices with Spring Boot and Spring Cloud** certification from **Udemy**.
- **PowerBI** certification from **DataCamp**.